InPlan — Detailed Class Diagram

Identity and roles
User
-id : UUID
-fullName : String
-email : String
-passwordHash : String
-active : Boolean
+authenticate(in password : String) : Boolean
+updateContact(in email : String) : void
+hasPermission(in action : String) : Boolean
Student
-studentNumber : String
-admissionYear : Integer
+createStudyPlan(in name : String) : StudyPlan
+viewMarks(in semester : String) : List<Mark>
+viewProgress(in specialization : Specialization) : AcademicProgress
AcademicOfficeEmployee
-employeeNumber : String
-officeName : String
+createProgramme(in name : String, in spec : Specialization) : ModelStudyProgramme
+requestReport(in field : FieldOfStudy) : AcademicReport
+requestReport(in spec : Specialization) : AcademicReport
Lecturer
-employeeNumber : String
-department : String
+canGrade(in course : Course) : Boolean
+assignMark(in enrollment : Enrollment, in value : Decimal) : Mark
+reviseMark(in mark : Mark, in value : Decimal) : void
Study planning and budget
StudyPlan
-id : UUID
-name : String
-status : String
-createdAt : DateTime
+selectSpecialization(in spec : Specialization) : void
+addCourse(in course : Course, in semester : String) : PlanItem
+removeCourse(in itemId : UUID) : void
+allocateBudget(in item : PlanItem, in amount : Decimal) : Boolean
+totalCredits() : Integer
+totalCost() : Decimal
+confirm(in actor : Student, out result : ValidationResult) : Boolean
PlanValidator
-checkPrerequisites : Boolean
+validate(in plan : StudyPlan, out result : ValidationResult) : Boolean
+reallocate(inout budget : CreditBudget, in delta : Decimal) : Boolean
PlanItem
-id : UUID
-semester : String
-allocatedAmount : Decimal
~setAllocation(in amount : Decimal) : void
+isFullyFunded(in cost : Decimal) : Boolean
ValidationResult
-errors : List<String>
-warnings : List<String>
-missingCoreCourses : Set<Course>
-missingCredits : Integer
+addError(in message : String) : void
+addWarning(in message : String) : void
+isValid() : Boolean
CreditBudget
-id : UUID
-limit : Decimal
-allocated : Decimal
+available() : Decimal
+tryChangeAllocation(in delta : Decimal) : Boolean
validates / reallocates
contains
fills out parameter
reads and updates
selects
owns
selected
confirmed as
Academic catalogue
FieldOfStudy
-id : UUID
-name : String
-description : String
+rename(in name : String) : void
+listCourses(in availableOnly : Boolean) : List<Course>
ModelStudyProgramme
-id : UUID
-name : String
-academicYear : String
+addCourse(in course : Course) : void
+removeCourse(in course : Course) : void
+compareWith(in plan : StudyPlan) : Set<Course>
Specialization
-id : UUID
-name : String
-requiredCredits : Integer
+addCoreCourse(in course : Course) : void
+missingCoreCourses(in selected : Set<Course>) : Set<Course>
+accepts(in course : Course) : Boolean
CourseCatalogue
-catalogueId : UUID
-updatedAt : DateTime
+findById(in courseId : UUID) : Course [0..1]
+filter(in field : FieldOfStudy [0..1], in credits : Integer [0..1], in spec : Specialization [0..1]) : List<Course>
Course
-id : UUID
-code : String
-title : String
-academicCredits : Integer
-cost : Decimal
-available : Boolean
+updateDetails(in title : String, in credits : Integer, in cost : Decimal) : void
+addPrerequisite(in prerequisite : Course) : void
+prerequisitesMet(in student : Student) : Boolean
classifies
groups
targets
includes
core courses
eligible courses
indexes
dependent courses
requires
prerequisites
requests progress
created by
requests report
academic history
authorized to teach / grade
last assigned / revised by
Academic records and reporting
ProgressService
-passThreshold : Decimal
+calculate(in student : Student, in spec : Specialization) : AcademicProgress
ReportService
-passThreshold : Decimal
+generateByField(in actor : AcademicOfficeEmployee, in field : FieldOfStudy) : AcademicReport
+generateBySpecialization(in actor : AcademicOfficeEmployee, in spec : Specialization) : AcademicReport
AcademicProgress
-earnedCredits : Integer
-requiredCredits : Integer
-missingCoreCourses : Set<Course>
+remainingCredits() : Integer
+completionPercentage() : Decimal
+isComplete(in requireAllCoreCourses : Boolean) : Boolean
Enrollment
-id : UUID
-semester : String
-confirmedAt : DateTime
-academicCreditsAtEnrollment : Integer
-costAtEnrollment : Decimal
+getMark() : Mark [0..1]
+earnedCredits(in passThreshold : Decimal) : Integer
AcademicReport
-id : UUID
-generatedAt : DateTime
-grouping : String
-groupId : UUID
-studentCount : Integer
-averageMark : Decimal [0..1]
-passRate : Decimal [0..1]
+export(in format : String) : ByteArray
Mark
-id : UUID
-value : Decimal
-assignedAt : DateTime
-updatedAt : DateTime
~revise(in value : Decimal, in actor : Lecturer) : void
+isPassing(in passThreshold : Decimal) : Boolean
creates
reads passed courses
reads academic records
creates
enrolled in
final result
1
0..*
0..1
1..*

Symbol	**Meaning**
+ / - / # / ~	public / private / protected / package
solid line + hollow triangle	generalization (inheritance)
solid line	association; numbers at both ends are multiplicities
filled diamond	composition; exclusive ownership of a part
hollow diamond	shared aggregation; independent part lifetime
dashed arrow	dependency; temporary use
in / out / inout	input / output / input and output parameter
0..1 / 1 / 0..* / 1..*	optional / exactly one / many / one or more

All amounts use the same budget unit; academic credits are separate.
Role subclasses implement User.hasPermission(action).
Domain constraints and design assumptions: task6.md.

Proposed design based on Lab 3 requirements and Lab 4 use cases